Consider a deep neural network with K convolution layers for 100 <= K <= 1000. Assess the performance in terms of speed-up – and wherever applicable – in terms of communication overhead, data movement overhead, and memory contention issues, where the software pipeline is realized using (i) a cluster of off-the-shelf computers, (ii) a multi-core CPU or (iii) a GPGPU system. **[9 Marks]**

In a deep neural network consisting of more than a hundred layers, it was observed that the edge device is not able to run. Likely cause was deemed to be lack of sufficient memory. Reducing the number of layers is not giving sufficient quality of result. Suggest methods to help overcome this hurdle, with analytical reasoning. If instead of running the network in a single processing unit it is possible to provide a 4-core unit with same amount of RAM but half the cache, which of the methods will still be relevant and why. **[6 Marks]**

The Government of Mogamba wants to detect any likely financial fraud by constantly monitoring transactions happening in any part of the country. They believe that all fraudulent transactions follow certain patterns that can be easily detected by just focusing on transactions targeting the same bank/financial institution. They called the country's Confederation of Banking and Financial Institutions, and asked them to come up with a solution. The confederation has assigned you the task of creating a robust ML system that meets the government's requirement without hurting their core business principles.

Describe – at a high level – the system required for a federated learning solution to meet the above requirement.
Mention the learning algorithm / solution used and explain the system with its components in detail.
Argue the scalability of your design and identify the factors it is dependent on. **[9 Marks]**

Consider the following operation.

$$\mathbf{A} + \mathbf{B} = (a_1 + b_1,\ a_2 + b_2,\ \ldots,\ a_n + b_n)$$

Evaluate the performance of a multi-threaded implementation of this operation, in multi-core CPUs and in GPUs – in terms of impact of caching, data movement overhead, and thread management overhead **[6 Marks]**